

LESSON 14.6

APPLYING PROPORTIONAL REASONING TO PROBABILITY



LESSON INTRODUCTION

MA.8.DP.2.3: Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

LEARNING TARGETS

- I can use proportional reasoning to make a prediction based on theoretical probability.
- I can use the probability of an experiment to determine the sample space.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

WORKING TOWARD

MA.9.12.DP.3.5 Solve real-world problems involving univariate and bivariate categorical data.

ESSENTIAL QUESTIONS

- How can sample spaces be determined from given probability?
- How can the validity of a probability distribution be verified?

LESSON NARRATIVE

In this lesson, students examine the validity of probability distributions. Students begin by comparing the probabilities of several decks of cards, understanding that the ratio of favorable outcomes to total outcomes remains the same, so does the probability. Students then apply what they have learned about the validity of probability distributions to determine the number of elements in a sample space and vice versa.

COMPONENT SUMMARY

14.6.1 WARM-UP (~5 MIN.)

Which spinner does not belong?

14.6.3 COLLABORATION (10 MIN.)

Find sample spaces given probability

14.6.5 TRY IT (10 MIN.)

Find probability, sample spaces, and probability distributions

14.6.7 WRAP-UP (~5 MIN.)

Analysis of likelihood of events and outcomes

14.6.2 EXPLORATION (10 MIN.)

Compare sample spaces and probability

14.6.4 BRING IT TOGETHER (5-10 MIN.)

Determine probability distributions

14.6.6 YOUR TURN! (5 MIN.)

Error Analysis and finding sample spaces

RESOURCES

GLOSSARY	MATERIALS	ADDITIONAL PREPARATION
Key Terms: Probability distribution	(Optional) Sentence stems	N/A
Additional Terms: Complement		

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not convert the whole number into a fraction correctly before multiplying it by a fraction.
- Students may confuse a larger sample space with a greater probability.

THINGS TO CONSIDER

- If time is an issue, then consider assigning 14.6.6 Your Turn! or 14.6.5 Try It components for homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

Question 2 in 14.6.3 Collaboration allows students to engage in a Compare and Connect routine. After students compare each other’s box of puzzle pieces, students can demonstrate and connect their strategies with how they satisfied the given condition.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

As students continue to engage with probability questions throughout this unit, they are building mathematical fluency with probability. They are offered multiple opportunities to practice efficient and generalizable methods. Monitor the ways students are using feedback from the teacher and peers to improve efficiency and accuracy when calculating probabilities.

DIFFERENTIATE INSTRUCTION

Lesson 6 provides a multitude of visual entry points for students as they work through probability questions in each section. 14.6.2 Exploration provides visual examples of decks of cards for students to utilize as they determine the probability of pulling a specific-colored card. Students also have the opportunity to find probability and compare patterns using proportional reasoning.

Additional differentiation opportunities are denoted throughout the Lesson Guide.



	Deck A (4 cards)	Deck B (8 cards)	Deck Y (36 cards)	Deck Z (192 cards)
Number of Purple Cards	1	2	9	48
Number of Green Cards	1	2	9	48
Number of Blue Cards	1	2	9	48
Number of Red Cards	1	2	9	48
P(not blue)	3	6	27	144

14.6.1 WARM-UP: PROPORTIONALITY

ENRICHMENT

Which One Doesn't Belong?

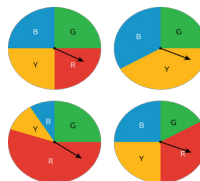
Each of the spinners does not belong for different reasons. Consider posting questions for students to think about when deciding which spinner does not belong, such as:

- What characteristics are you comparing to decide which spinner does not belong?
- How are you going to communicate which spinner you chose that does not belong?



14.6.1 Warm-Up: Proportionality

1. Four spinners are shown.



Explain which spinner doesn't belong.

Answers will vary. All answers are possible. For example, a possible answer could be spinner two because it is the only spinner that does not have four colors.



14.6.2 Exploration: Increasing Sample Space

Work with your partner to complete the following.

1. A deck of cards is shown.

Deck A:



If a game is played in which a card is drawn from Deck A, what is the probability of drawing a card that is **not** blue?

$\frac{3}{4}$ or 0.75 or 75%

2. Consider Deck B.

Deck B:



- a. What is the probability of drawing a card that is **not** blue from Deck B?

$\frac{6}{8}$ or 0.75 or 75%

- b. Compare this to P(not blue) in Deck A. List the similarities and differences in these experiments.

Answers will vary. The probabilities are equivalent, there are just more cards in Deck B. The ratio of cards that are not blue to cards is equivalent for both decks.

- c. If Deck B is doubled and there were 4 cards of each color, explain why or why not P(not blue) would change.

The probabilities will not change because there is still going to be the same ratio of blue cards to not blue cards.

3. Now, consider Deck C.

Deck C:



Explain why P(not blue) is different for Deck C and Deck B.

The probability of not blue in Deck C is different because there is one less blue card compared to the total, making the probability $\frac{1}{8}$ instead of $\frac{2}{8}$.

4. Complete the table below by increasing the size of Deck A without changing the probabilities of any colors.

	Deck A (4 cards)	Deck B (8 cards)	Deck Y (36 cards)	Deck Z (192 cards)
Number of Purple Cards	1	2	9	48
Number of Green Cards	1	2	9	48
Number of Blue Cards	1	2	9	48
Number of Red Cards	1	2	9	48
P(not blue)	$\frac{3}{4}$	$\frac{6}{8}$	$\frac{27}{36}$	$\frac{144}{192}$

14.6.2 EXPLORATION: INCREASING SAMPLE SPACE

TEACHER MOVES

This Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the 14.6.4 Bring It Together component. Students may revise their thinking throughout the Exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group Bring It Together.

14.6.2 EXPLORATION: INCREASING SAMPLE SPACE (CONTINUED)

5. Summarize the strategy you used to complete the table.

Answers will vary. All card colors have the same ratio of cards. In each deck, $\frac{3}{4}$ of the total cards are not blue. To determine the number of each color in the decks or the total number of cards in each deck, I found the scale factor to multiply the values in Deck A. All values are multiplied by 2 for Deck B, by 9 for Deck Y, and by 48 for Deck Z.

14.6.3 COLLABORATION: PUZZLE PROPORTIONS

DIFFERENTIATE INSTRUCTION

Prior to completing question 2c, provide sentence stems to assist students with demonstrating how their partner's box satisfies or does not satisfy the condition, such as:

- My partner's box contains _____, so it satisfies the condition because _____.
- My partner's box is missing _____, which means that it does not satisfy the condition because _____.

TEACHER MOVES

Observe and listen as students work with their partners on question 4. Make note of student ideas that you can ask them to share in the whole group 14.6.4 Bring It Together.



14.6.3 Collaboration: Puzzle Proportions

Souma is putting together a puzzle that contains white, red, and yellow pieces. He knows that the probability of pulling a white puzzle piece out of the box is $\frac{1}{3}$.

1. Draw the box of puzzle pieces as you think it would exist. Make sure to label the colors of your pieces.
Answers will vary.
2. Compare your box of puzzle pieces with your partner's.
 - a. How many pieces did your partner draw?
Answers will vary.
30 pieces.
 - b. How many of those were white puzzle pieces?
Answers will vary.
10 pieces.
 - c. Demonstrate how their box satisfies (or does not satisfy) the condition that the probability of pulling a white puzzle piece out of the box is $\frac{1}{3}$.
Answers will vary.
 $\frac{10}{30} = \frac{1}{3}$
3. Make any necessary changes to your box of puzzle pieces based on feedback from your partner.
4. Souma counts his puzzle pieces and determines that there are 60 pieces. If $P(\text{white}) = \frac{1}{3}$ and $P(\text{red}) = 0.6$, then ...
 - a. ...how many red pieces are in his puzzle box?
36 red pieces
 - b. ...how many yellow pieces are in his puzzle box?
4 yellow pieces



14.6.4 Bring It Together: Probability Distributions



For a **probability distribution** to be valid, each probability must be a number between 0 and 1 and the sum of each outcome's probabilities must equal 1.

1. Complete the following probability distribution so that it is valid.

Selecting clothing items from a hamper of mixed items that includes dresses, t-shirts, leggings, and shorts containing 30 items:

	Dresses	T-Shirts	Leggings	Shorts	Total
Frequency	10	9	5	6	30
Probability	$\frac{1}{3}$	0.3	$\frac{1}{6}$	20%	100%

2. Zareen has a bag of 10 sour candies. The bag consists of orange, yellow, green, blue, and red candies. Complete the table below by using probability to determine how many candies of each color are in Zareen's bag.

	Orange	Yellow	Green	Blue	Red
Frequency	2	4	1	2	1
Probability	0.2	40%	0.1	$\frac{1}{5}$	10%

14.6.4 BRING IT TOGETHER: PROBABILITY DISTRIBUTIONS

TEACHER MOVES

This component is intended to be teacher-led. It provides an opportunity to bring together ideas from the previous two components. Have students share strategies based on thinking observed during each component. Also use this component to address any common misconceptions observed and solidify understanding of how to apply proportional reasoning to probability distributions.

TEACHER MOVES

For question 1, model for students how to complete the probability distribution using proportional reasoning and the information given. For question 2, consider releasing students to try this with a partner before going over it with the whole group.

14.6.5 TRY IT: DETERMINING SAMPLE SPACE

DIFFERENTIATE INSTRUCTION

In order to assist students with completing question 1a, provide sentence starters, such as:

- First, Ziggy can _____.
- Next, Ziggy can _____.
- Ziggy could also _____.
- Ziggy should _____ and then should _____.



14.6.5 Try It: Determining Sample Space

1. Ziggy made a dozen chocolate chip cookies for his family. Each cookie either had 9, 10, or 11 chocolate chips. Before baking them, he counted how many chocolate chips were in each cookie and determined the following:

$$P(\text{picking a cookie with 9 chocolate chips}) = \frac{1}{6}$$

$$P(\text{picking a cookie with 10 chocolate chips}) = 0.5$$

$$P(\text{picking a cookie with 11 chocolate chips}) =$$

- a. Ziggy forgot to calculate the probability of selecting a cookie with 11 chocolate chips. Explain how he can use the two probabilities he already calculated to determine the missing probability.

Answers will vary. The probability of all events has to sum to 100% or 1.0. Two cookies 9 chocolate chips ($2 \cdot \frac{1}{6}$). Six cookies have 10 chocolate chips, ($6 \cdot 0.5$). That means the rest of the 12 cookies have 11 chips, or $\frac{4}{12}$ cookies have 11 chips, for a probability of $\frac{1}{3}$.

- b. What is the probability of choosing a cookie with 11 chocolate chips?

$$\frac{1}{3}$$

- c. How many cookies have exactly 9 chocolate chips?

Two cookies

- d. How many cookies have exactly 10 chocolate chips?

Six cookies

- e. How many cookies have exactly 11 chocolate chips?

Four cookies

- f. Construct a probability distribution to check your work.

Number of Chocolate Chips	Probability
9	0.16
10	0.5
11	0.3



14.6.6 Your Turn!

1. A grocery store is giving away gift cards to customers who can correctly estimate the number of green apples in a bag of 25 apples. The sign states the following:

$$P(\text{Granny Smith}) = 0.2$$



$$P(\text{Honeycrisp}) = 0.32$$



$$P(\text{Red Delicious}) = 0.48$$



Aubree says, "Since there are 25 apples in the bag and Granny Smith is the only type that is green, we must multiply the probability of selecting a Granny Smith apple by the total number of apples. So, there are 5 green apples since $0.2 \cdot 25 = 5$."

Joey states, "I don't think that's correct. There could also be Crispin apples in the bag. Those are green as well."

Aubree says, "The sign only mentions Granny Smith, Honeycrisp, and Red Delicious. I don't think that there are other apples in the bag."

Joey says, "Yes, the sign only mentions those three, but this probability distribution is not valid. When I add the probabilities, I get 0.82, which is not 1. Therefore, there must be another type of apple in the bag."

14.6.6 YOUR TURN!

TEACHER MOVES

If time is an issue, then consider assigning 14.6.6 Your Turn! for homework.

ENRICHMENT

Prior to answering question 1a or performing any calculations, ask students to estimate the number of green apples given only the information on the sign. After students complete the question, encourage them to compare their estimation to their calculation to see if they would have won the gift card.

14.6.6 YOUR TURN! (CONTINUED)

TEACHER MOVES

Students will calculate the number of each type of apple in the bag. Discuss different strategies that students can use to perform the calculations. Students may state that they can multiply the probability given by the total number of apples. Students may also suggest converting the decimals into equivalent fractions and finding the fraction of the total for each apple.

- a. Explain who made an error in their calculations.
Joey added the probabilities incorrectly. The probabilities do equal one.
- b. What is the correct number of green apples in the bag?
5 green apples
- c. Calculate the number of Honeycrisp apples in the bag.
8 Honeycrisp apples
- d. Calculate the number of Red Delicious apples in the bag.
12 Red Delicious apples

**14.6.7 Wrap-Up: Cool Down**

Consider the following three events.

- A. The probability of spinning 1 on the spinner is 50%.
- B. The probability of spinning 2 on the spinner is 0.125.
- C. The probability of spinning 3 on the spinner is $\frac{3}{8}$.

1. Which event is the most likely?

Event A

2. Explain whether or not there could be another outcome on the spinner.

There could not be another option because the probabilities equal 1.

14.6.7 WRAP-UP: COOL DOWN**DIFFERENTIATE INSTRUCTION**

Consider providing students with an image of a spinner to assist them with completing the Wrap-Up. A visual will provide students with an alternate strategy to connect the events with their probability.

!!!